

How much leaf is left?

A loop goes somewhere. · Curiosity day

Name _____

Date _____



A drawing model for a journey. Use the invented table distances, not the picture's size or dot spacing.

PRETEND JOURNEY DISTANCES · MILLIMETERS

Part	Whole route	First stretch	Next stretch
Length (mm)	32	12	9

A pretend journey: mm means millimeters. Use these numbers; do not measure the painting or assume one loop equals one inch.

1. The inchworm travels 12 mm, then 9 mm. How far has it traveled in all?

$$12 + 9 = \underline{\hspace{2cm}}$$

Draw or write your thinking.

2. The whole route is 32 mm. After the 12 mm and 9 mm stretches, how far is left? Show both steps.

$$32 - (12 + 9) = \underline{\hspace{2cm}}$$

Draw or write your thinking.

3. Extend the 32 mm route by 8 mm. After traveling the same 21 mm, how far is left on this longer route?

$$(32 + 8) - 21 = \underline{\hspace{2cm}}$$

Draw or write your thinking.

Explain why adding 8 mm to the route also adds 8 mm to the distance left, if the traveled distance stays the same.

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A green arch rises on a leaf. "A tiny bridge!" says Little. Then the arch flattens and the head reaches ahead. Big watches beside the path. This is an inchworm, a kind of caterpillar. How can bending help it get somewhere?



The front legs grip the leaf. The rear comes closer, lifting the long middle into a loop. Near the rear are fleshy grips called prolegs. "No walking legs under that middle!" Little notices. With its ends closer, the body must bend.



Now the prolegs anchor the rear: they hold it in place. The front reaches ahead as the body straightens. Once the front grips, the rear can catch up again. "One end holds while the other moves," Little says. A loop becomes a journey.



Little sketches the head before a vein fork, then beyond it. The changed positions are evidence: clues that can support an idea. "It traveled past the fork!" That is an inference, an idea drawn from clues. The same leaf vein makes the comparison fair.



"Does every loop go one inch?" Little asks. The name sounds like measuring, but real steps vary. Little draws a pretend route instead. Its distances are invented. A model helps answer a math question without pretending the leaf picture is a ruler.



Little imagines a later chapter. The caterpillar becomes a pupa, then changes into a moth. This change of body form is metamorphosis. It takes time. Little sketches the stages. "A new shape after all those loops!"

Think about it: What evidence shows the head traveled rather than only bending in place? Why compare it with the same vein fork? During the reach, what anchors the rear? Predict what could happen if that grip slipped, and explain why.

Explain a tiny grip.

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Choose a scene. Write two linked sentences: one observation, then an explanation using because. Draw a detail that supports your idea.



The rear catches up



A grip, then a reach



The leaf keeps a clue

Draw the detail you noticed. Add an arrow to show movement or a dot to show position.

prolegs · anchor · loop · reach · evidence · inference · because

The middle rises because ... What happens when the rear gets closer to the gripping front?

The rear grip matters because ... What keeps the body in place while the front reaches?

I infer that it moved because ... Compare the head with the same vein fork in both sketches.

Reread your words. Does the detail support your explanation? Check the first capital and each ending mark.